SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Machine Learning

ASSESSMENT 1 – Analytical Problem solving (5 QUESTIONS)

Course Code:	ITA05	**Course Title:**	**Computer Vision**
CO Assessed:	CO2	**Assessment:**	Assessment 1: Analytical Problem solving
Weightage:	35% of CIA	**Total Marks:**	(5 Questions × 10 Marks)
CO2 Statement:	***Apply image enhancement, filtering, and segmentation techniques to analyze images, detect edges and boundaries, and solve real-world Computer Vision problems.***		
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Section / Batch:	BE - CSE (FINAL YEAR)	**Date:**	21/10/2026

Code of Conduct

I Maredi Neela Meghana - 192311002 (NAME/ REG NO) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: **MAREDI NEELA MEGHANA**

QUESTIONS: 5 **Total Marks: 50**

Q1. A grayscale image contains the following pixel intensity values: 10, 10, 20, 20, 20, 30, 30, and 40. Construct the histogram by calculating the frequency of each intensity value. Calculate the cumulative distribution function (CDF) for the given image. Apply histogram equalization using the formula s = round((L − 1) × CDF), where L = 256. Determine the new intensity values and compare the contrast distribution before and after equalization.

Answer

Histogram equalization is an image-enhancement technique used to improve contrast. It redistributes intensity values so that the available gray levels are used more effectively. In an 8-bit grayscale image, intensities range from 0 to 255, so L = 256 and L − 1 = 255.

1. Given data and histogram

The eight pixel values are 10, 10, 20, 20, 20, 30, 30, and 40. The frequency is the number of times each distinct intensity occurs. The total number of pixels is N = 8.

Intensity (r)	**Frequency (n_r)**	**Probability n_r/N**	**Cumulative frequency**	**CDF**
10	2	2/8 = 0.250	2	2/8 = 0.250
20	3	3/8 = 0.375	5	5/8 = 0.625
30	2	2/8 = 0.250	7	7/8 = 0.875
40	1	1/8 = 0.125	8	8/8 = 1.000
Total	8	1.000	—	—

The histogram therefore has four non-zero bars: intensity 10 has frequency 2, intensity 20 has frequency 3, intensity 30 has frequency 2, and intensity 40 has frequency 1. All other intensity levels have frequency zero.

2. Calculate the cumulative distribution function (CDF)

The CDF at an intensity is the cumulative number of pixels up to and including that intensity divided by the total number of pixels. It is calculated by adding the probabilities from the lowest intensity to the current intensity.

For intensity 10: CDF = 2/8 = 0.25. For intensity 20: CDF = (2 + 3)/8 = 5/8 = 0.625. For intensity 30: CDF = (2 + 3 + 2)/8 = 7/8 = 0.875. For intensity 40: CDF = 8/8 = 1.00.

3. Apply histogram equalization

Use the given formula s = round((L − 1) × CDF) = round(255 × CDF). Each original intensity is replaced by the output intensity calculated from its CDF.

Original intensity	CDF	Calculation	New intensity s
10	0.250	round(255 × 0.250) = round(63.75)	64
20	0.625	round(255 × 0.625) = round(159.375)	159
30	0.875	round(255 × 0.875) = round(223.125)	223
40	1.000	round(255 × 1.000) = round(255)	255

Thus, the original sequence [10, 10, 20, 20, 20, 30, 30, 40] becomes [64, 64, 159, 159, 159, 223, 223, 255]. The frequency of each group stays the same, but the intensity levels assigned to those pixels change.

4. Contrast comparison

Feature	Before equalization	After equalization
Minimum intensity	10	64
Maximum intensity	40	255
Intensity range	40 − 10 = 30	255 − 64 = 191
Distribution	Values concentrated in a narrow dark range	Values spread across a wider range

Before equalization, all pixels lie between 10 and 40, so the image uses only a small part of the available 0–255 range. It may look dark or have low contrast. After equalization, the values are spread from 64 to 255. The larger range can make differences between gray levels easier to see and can reveal details that were difficult to distinguish.

Conclusion: The equalized values are 64, 159, 223, and 255 for original intensities 10, 20, 30, and 40 respectively. Histogram equalization improves the spread of contrast in this example. The exact visual improvement depends on the image; equalization can sometimes make noise or unwanted variations more noticeable.

Q2. A 3 × 3 image has the intensity values [[10, 20, 30], [20, 100, 20], [10, 20, 30]]. Apply a 3 × 3 averaging filter to calculate the new intensity value of the centre pixel. Replace the centre pixel with the calculated average while keeping the other pixels unchanged. Apply a median filter to the same neighbourhood and calculate the centre pixel value. Compare the results of the averaging and median filters and determine which filter better preserves image details in the presence of noise.

Answer

Spatial filtering changes a pixel using the values in its local neighbourhood. Here, the centre pixel is 100 and the surrounding pixels are mostly between 10 and 30. The high centre value may represent a bright detail or noise. An averaging filter and a median filter process the same 3 × 3 neighbourhood in different ways.

1. Given image

10	**20**	**30**
20	100	20
10	20	30

The nine values in the neighbourhood are 10, 20, 30, 20, 100, 20, 10, 20, and 30.

2. Apply the 3 × 3 averaging filter

An averaging filter replaces the centre pixel with the arithmetic mean of all nine pixels. For a 3 × 3 filter, add the nine values and divide by 9.

Sum = 10 + 20 + 30 + 20 + 100 + 20 + 10 + 20 + 30 = 260

Average = 260/9 = 28.89 ≈ 29

Therefore, the new centre pixel value is approximately 29. Keeping all other pixels unchanged, the filtered image is:

10	**20**	**30**
20	29	20
10	20	30

The average reduces the original centre value from 100 to about 29. However, the centre value is influenced by the high value 100, which pulls the average upward compared with the other surrounding values.

3. Apply the median filter

A median filter sorts the nine neighbourhood values from smallest to largest and selects the middle value. It is less affected by a single unusually high or low pixel than the arithmetic mean.

Sorted values: 10, 10, 20, 20, 20, 20, 30, 30, 100

There are nine values, so the median is the fifth value. The fifth value is 20. Therefore, the new centre pixel value after median filtering is 20.

Keeping the other pixels unchanged, the median-filtered image is:

10	**20**	**30**
20	20	20
10	20	30

4. Comparison of the two filters

Feature	Averaging filter	Median filter
Method	Calculates the mean	Sorts values and selects the middle value
Centre output	28.89, approximately 29	20
Effect of centre value 100	Influences the mean strongly	Has little influence on the median
Noise removal	Can reduce random noise but may blur edges	Effective for isolated impulse-like noise
Detail preservation	May smooth fine edges and details	Usually preserves edges better

5. Which filter is better in the presence of noise?

For this neighbourhood, the median filter is the better choice if the value 100 is an unwanted isolated noise pixel. It replaces that outlier with 20, which is consistent with the surrounding values. The averaging filter gives about 29 because the large value 100 raises the mean. It also tends to blur boundaries because it mixes values from both sides of an edge.

Averaging is useful when the goal is general smoothing and the noise is spread across many pixels. Median filtering is especially useful for salt-and-pepper noise, where isolated pixels become very bright or very dark. Neither filter is best for every image: the correct choice depends on the noise type and whether preserving sharp boundaries is important.

Conclusion: The averaging-filter centre value is $28.89 \approx 29$, while the median-filter centre value is 20. The median filter better preserves local image details when an isolated extreme value is noise.

Q3. Consider a grayscale image with the intensity matrix [[50, 50, 50], [50, 100, 50], [50, 50, 50]]. Apply the Laplacian filter mask [[0, −1, 0], [−1, 4, −1], [0, −1, 0]] to calculate the response at the centre pixel. Use the calculated response to sharpen the centre pixel by adding it to the original intensity value. Determine the resulting centre pixel intensity after sharpening. Explain how the Laplacian filter enhances intensity transitions and fine details in the image.

Answer

The Laplacian filter is a second-order derivative filter used to highlight rapid changes in image intensity. These changes often occur at edges and fine details. In this question, the given mask has a positive coefficient at the centre and negative coefficients at its four direct neighbours.

1. Given image and mask

Image:

50	**50**	**50**
50	100	50
50	50	50

Laplacian mask:

0	**−1**	**0**
−1	4	−1
0	−1	0

2. Calculate the Laplacian response

Multiply each image pixel by the corresponding mask value and add the products. The four diagonal pixels do not contribute because their mask coefficients are zero.

Response = (0 × 50) + (−1 × 50) + (0 × 50) + (−1 × 50) + (4 × 100) + (−1 × 50) + (0 × 50) + (−1 × 50) + (0 × 50)

Response = −50 − 50 + 400 − 50 − 50 = 200

Therefore, the Laplacian response at the centre pixel is 200. The positive response occurs because the centre value 100 is much brighter than the four neighbouring values, which are all 50.

3. Sharpen the centre pixel

The question specifies that the Laplacian response must be added to the original centre intensity.

Sharpened centre intensity = Original centre intensity + Laplacian response

= 100 + 200 = 300

The mathematical result is 300. However, a standard 8-bit grayscale image can store only intensities from 0 to 255. If the result is stored directly in an 8-bit image, it must be clipped to 255 (or handled by suitable rescaling). Thus, the calculated sharpened value is 300, and its valid displayed 8-bit value after clipping is 255.

4. How the Laplacian enhances edges and fine details

In a region where neighbouring pixels have similar intensities, the weighted sum tends to be small. At an edge, the centre pixel may differ strongly from its neighbours, producing a larger positive or negative response. This makes sudden intensity transitions stand out. Fine details, such as thin lines or small bright spots, can also create rapid changes and therefore receive a strong response.

For sharpening, the Laplacian response is combined with the original image. With the mask used here, the question explicitly directs us to add the response. This increases the centre value where the response is positive, making the bright centre stand out more clearly from its surroundings. Other Laplacian mask conventions use the opposite sign and therefore subtract the response; the sign must always match the chosen mask.

5. Advantages and limitation

An advantage of Laplacian sharpening is that it is simple and can enhance small details without needing to identify a specific edge direction. A limitation is that derivative filters are sensitive to noise, because noise also causes abrupt intensity changes. Strong responses can therefore appear around unwanted noise. In practical systems, smoothing may be applied before sharpening, and output values should be clipped or normalized carefully.

Conclusion: The Laplacian response is 200. Adding it to the original centre value gives 300 mathematically; for an 8-bit image, the value is clipped to 255. The filter emphasizes intensity transitions and fine details, helping the image appear sharper.

Q4. A grayscale image contains the pixel intensity values [[40, 60, 120], [80, 150, 200], [30, 100, 180]]. Apply global thresholding with a threshold value of 100, assigning 1 to pixels greater than or equal to the threshold and 0 to the remaining pixels. Construct the resulting binary image. Calculate the number of foreground and background pixels after thresholding. Explain how the selected threshold separates objects from the background and identify one limitation of this method.

Answer

Global thresholding is a simple image-segmentation method that converts a grayscale image into a binary image. A single threshold is applied to every pixel. In this problem, the threshold is T = 100. Each pixel with intensity greater than or equal to 100 is assigned 1, while each pixel below 100 is assigned 0.

1. Given grayscale image

40	**60**	**120**
80	150	200
30	100	180

The matrix contains nine pixels. We examine each value and compare it with the threshold of 100.

2. Apply the threshold rule

Rule: if pixel intensity ≥ 100, output 1; otherwise, output 0.

Original row	**Threshold decisions**	**Binary row**
40, 60, 120	0, 0, 1	[0, 0, 1]
80, 150, 200	0, 1, 1	[0, 1, 1]
30, 100, 180	0, 1, 1	[0, 1, 1]

The resulting binary image is:

0	**0**	**1**
0	1	1
0	1	1

3. Count foreground and background pixels

Foreground pixels are represented by 1. They correspond to original intensities 120, 150, 200, 100, and 180.

Number of foreground pixels = 5

Background pixels are represented by 0. They correspond to original intensities 40, 60, 80, and 30.

Number of background pixels = 4

Pixel class	**Binary value**	**Count**	**Original intensities**
Foreground	1	5	120, 150, 200, 100, 180
Background	0	4	40, 60, 80, 30
Total	—	9	All pixels

4. How the threshold separates object and background

The threshold separates pixels according to brightness. Pixels at or above 100 are treated as the object or foreground, and pixels below 100 are treated as the background. This can be effective when the object is consistently brighter than its surroundings. The binary result is simpler to process than the original grayscale image and can support later tasks such as counting objects, measuring areas, or finding shapes.

The threshold value strongly affects the result. If the threshold is set too low, many background pixels may become foreground. If it is too high, some object pixels may be incorrectly removed. In this example, the value 100 is included in the foreground because the stated rule uses greater than or equal to the threshold.

5. One limitation of global thresholding

A major limitation is that one threshold may not work well when illumination varies across an image. For example, an object in a shadow may have lower intensities than the same object in a bright area. A single threshold could classify part of the object as background. Similar problems occur when the foreground and background have overlapping intensity values. Adaptive or local thresholding can help by using different thresholds for different image regions.

Conclusion: The binary image is [[0, 0, 1], [0, 1, 1], [0, 1, 1]]. It contains 5 foreground pixels and 4 background pixels. Global thresholding is simple and fast, but its performance depends on choosing a suitable threshold and having reasonably consistent lighting.

Q5. A grayscale image has the intensity matrix [[10, 10, 10], [10, 50, 90], [10, 50, 90]]. Apply the Sobel operator using the masks Gx = [[−1, 0, 1], [−2, 0, 2], [−1, 0, 1]] and Gy = [[−1, −2, −1], [0, 0, 0], [1, 2, 1]] to calculate the gradient components at the centre pixel. Calculate the gradient magnitude using $G = \sqrt{(Gx^2 + Gy^2)}$. Determine the dominant edge direction based on the calculated gradient components. Explain how the detected edge information can be used for boundary detection and image segmentation.

Answer

The Sobel operator detects intensity changes and estimates the image gradient in horizontal and vertical directions. It uses two 3 × 3 masks: Gx responds mainly to changes from left to right, and Gy responds mainly to changes from top to bottom. The gradient magnitude indicates the strength of the change, while the component values help determine its direction.

1. Given image and Sobel masks

Image:

10	**10**	**10**
10	50	90
10	50	90

Horizontal-gradient mask Gx:

−1	**0**	**1**
−2	0	2
−1	0	1

Vertical-gradient mask Gy:

−1	**−2**	**−1**
0	0	0
1	2	1

2. Calculate Gx at the centre

Multiply each pixel by the matching Gx coefficient and add the products.

Gx = [(-1 × 10) + (0 × 10) + (1 × 10)] + [(-2 × 10) + (0 × 50) + (2 × 90)] + [(-1 × 10) + (0 × 50) + (1 × 90)]

Gx = (−10 + 10) + (−20 + 180) + (−10 + 90)

$Gx = 0 + 160 + 80 = 240$

Therefore, the horizontal gradient component is Gx = 240. The right side of the neighbourhood is brighter than the left side, producing a strong positive response.

3. Calculate Gy at the centre

Apply the Gy mask in the same way.

$Gy = [(-1 \times 10) + (-2 \times 10) + (-1 \times 10)] + [(0 \times 10) + (0 \times 50) + (0 \times 90)] + [(1 \times 10) + (2 \times 50) + (1 \times 90)]$

$Gy = (-10 - 20 - 10) + 0 + (10 + 100 + 90)$

$Gy = -40 + 200 = 160$

Therefore, the vertical gradient component is Gy = 160. This positive value indicates that intensity also increases from the upper part of the neighbourhood toward the lower part.

4. Calculate gradient magnitude

The gradient magnitude combines the horizontal and vertical components:

$G = \sqrt{(Gx^2 + Gy^2)}$

$G = \sqrt{(240^2 + 160^2)} = \sqrt{(57{,}600 + 25{,}600)} = \sqrt{83{,}200}$

$G \approx 288.44$

The gradient magnitude is approximately 288.44. A larger magnitude indicates a stronger intensity change and therefore a more likely edge location.

5. Determine the dominant edge direction

Since |Gx| = 240 is greater than |Gy| = 160, the horizontal gradient component is stronger. This means the intensity changes more strongly from left to right. Such a change commonly indicates a predominantly vertical edge, because the edge itself runs approximately perpendicular to the gradient.

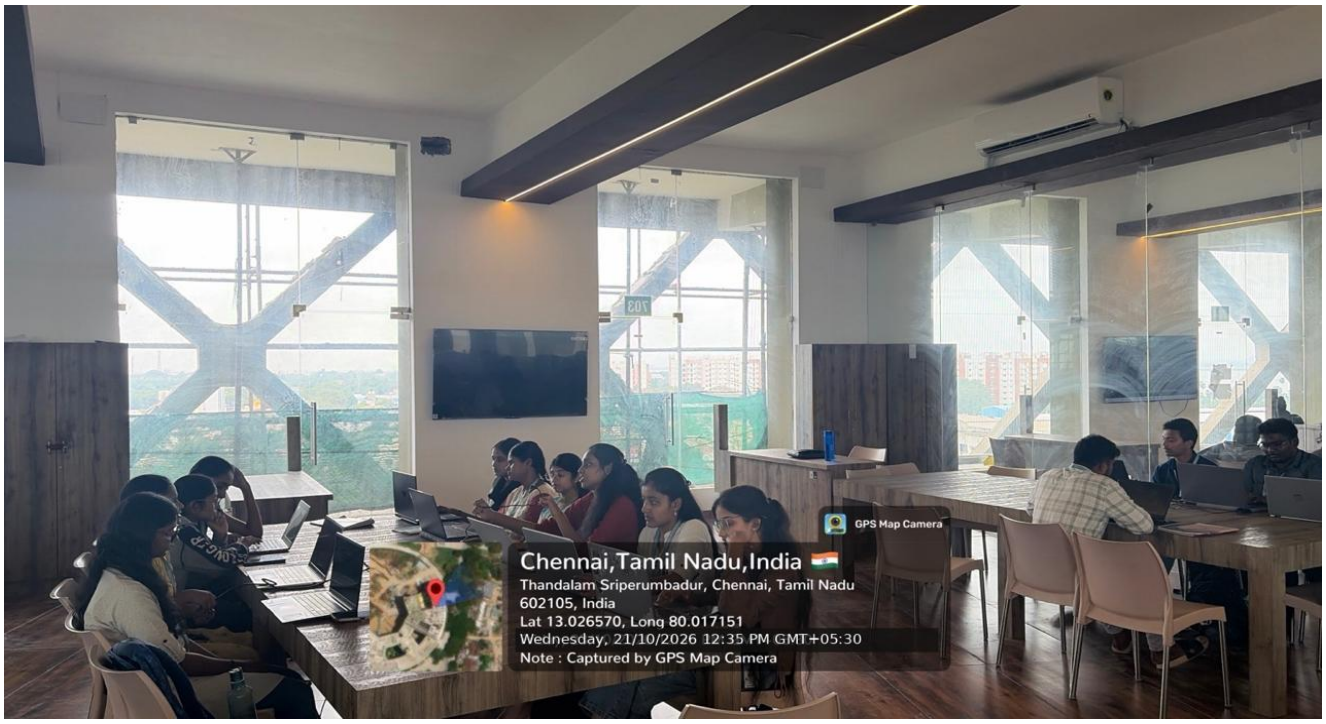
The gradient angle measured from the positive horizontal axis is $\theta = \tan^{-1}(Gy/Gx) = \tan^{-1}(160/240) \approx 33.7°$. This is the direction of greatest intensity increase, not the direction along the edge. The approximate edge orientation is perpendicular to this gradient, about 123.7° (equivalently, an undirected edge angle of about −56.3°). For a simple qualitative answer, the dominant change is horizontal and the corresponding edge is mainly vertical.

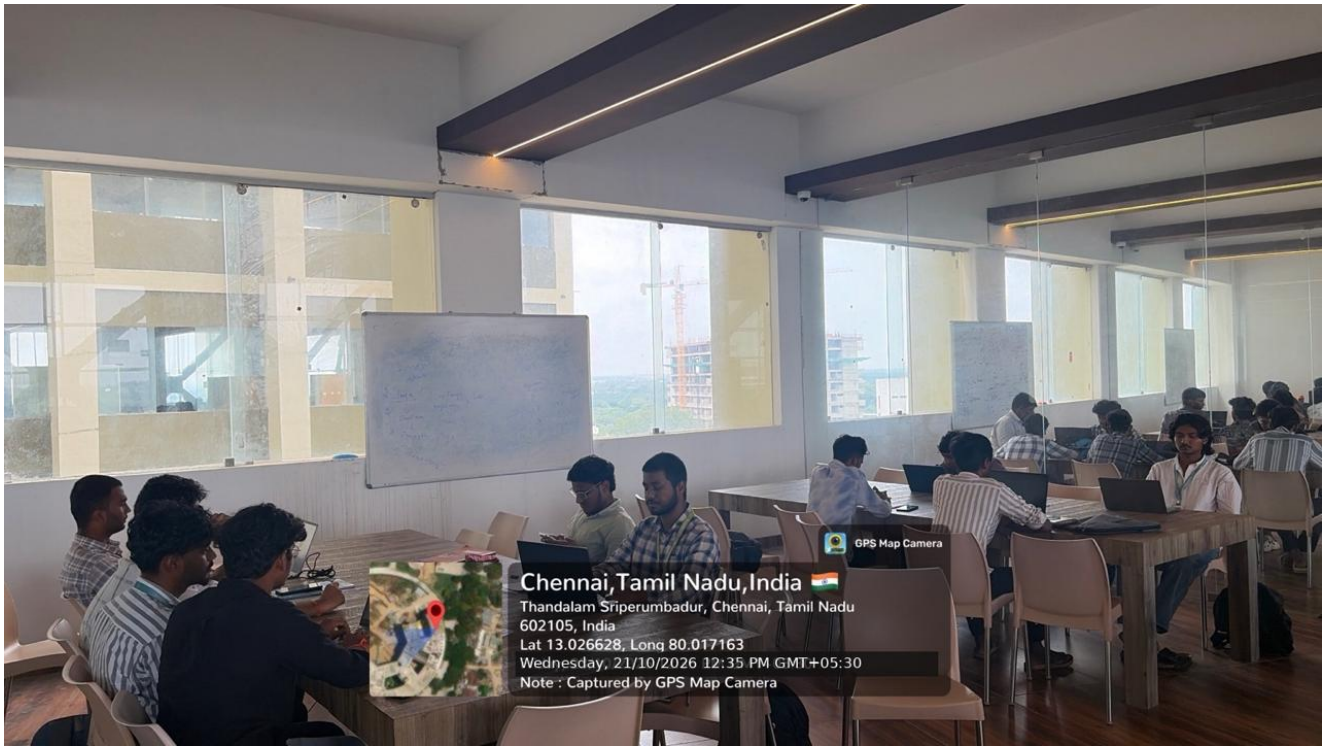
6. Use in boundary detection and segmentation

Sobel edge information can be used to locate boundaries between regions with different intensities. A system may calculate the gradient magnitude at each pixel and mark pixels whose magnitude exceeds a chosen threshold as edge pixels. These edges can help outline objects, identify shapes, or separate neighbouring regions.

In image segmentation, edge maps can be combined with thresholding, region-based methods, or contour detection. For example, a boundary around an object in a medical or industrial image can help estimate its size or distinguish it from the background. Sobel is computationally simple and also gives directional information. However, weak edges may be missed, and noise can create false edges; smoothing and suitable threshold selection can improve the result.

Conclusion: Gx = 240, Gy = 160, and the gradient magnitude is approximately 288.44. The stronger Gx component indicates a stronger left-to-right intensity change, suggesting a mainly vertical edge. Sobel gradients are useful for finding boundaries and supporting image segmentation.





The following assessment has been discussed by the faculty and done at the class on 21st October 2026.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Content Accuracy	30	Highly accurate, indepth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Faculty In-charge	Course Coordinator	Program Director
Signature: _____ Date: _____	Signature: _____ Date: _____	Signature: _____ Date: _____